

## Packet Analysis, Network Enumeration, Linux Networking, and Multi-Stage Lab

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts is 2 hours and 15 minutes.

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### Kali Networking + Linux Services

Start off with the command: **ip addr**. Then, **ip route**. The **ip route** command is used to view and manage the routing table of the system. Reminder: the routing table tells your computer: where to send network traffic, which gateway/router to use, and which network interface to send packets through.

Now, **ss -tulnp**. Next, **hostname**. The **hostname** command displays or changes the system's hostname. A hostname is assigned to a computer on a network.

What is the loopback interface?

*A virtual network interface a computer uses to communicate with itself. It does not connect to any physical network hardware. The most common loopback IP address is: **127.0.0.1**.*

*Commonly called: **localhost**.*

What is a default gateway in Linux?

*The router or device that your computer sends traffic to when the destination is outside the local network.*

What are listening sockets?

*A network port on a computer that is waiting for incoming connections. When a program wants to accept network connections, it opens a socket and puts it into a **LISTEN** state.*

To find IP address and gateway on Kali: just run **ip addr**. To find listening ports: **ss -tulnp**. In order to find active services, run the command **systemctl list-units --type=service --state=running**

For services in general: run **systemctl list-units --type=service**

What is systemd?

*The main init system and service manager used by most modern Linux distributions. It is responsible for: starting the OS during boot, launching and managing services/processes, handling background daemons, and managing logs, networking, timers, and more.*

What is systemctl?

*The command-line tool used to interact with systemd.*

Why are services important in Linux?

*They allow the OS to run essential background tasks automatically without user interaction. A service (also called a daemon) is a program that runs in the background and provides functionality to the system or network.*

Now, ping [google.com](https://www.google.com). Then, traceroute [google.com](https://www.google.com).

What is TTL in networking?

*Stands for Time To Live in networking. It is a value inside an IP packet that limits how long the packet can travel across networks. Every time a packet passes through a router: the TTL value decreases by 1. If TTL reaches: 0, the router discards the packet.*

Why does traceroute work?

*It works by using TTL field in IP packets to discover each router (hop) along the path to a destination. The idea is: send packets with gradually increasing TTL values and watch which routers respond.*

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## Wireshark Deep Packet Analysis

Capture Traffic, Browse Websites, ping devices, run nslookup, run Nmap scans, refresh Discord/browser traffic.

Filter: **icmp**.

What is ICMP?

*Stands for Internet Control Message Protocol. A network protocol used for: error reporting, diagnostics, and network communication feedback.*

What fields exist inside an ICMP packet?

*Type: what kind of ICMP message it is*

*Code: more specific detail about the type*

*Checksum: error-checking/integrity validation*

*Data: extra information/payload*

Filter: **dns**

What is an A record?

*Address Record is a type of DNS record that maps a domain name to an IPv4 address.*

What is an AAAA record?

*DNS record that maps a domain name to an IPv6 address.*

Filter: **tcp**

What is a TCP retransmission?

*When TCP resends a packet because it believes the original packet was lost or never received correctly.*

*Step 1: Sender sends data. Seq = 1000.*

*Step 2: Receiver should respond. ACK = 1001*

*Step 3: ACK never arrives.*

*Step 4: TCP retransmits. It resends the same packet. Seq = 1000.*

What causes packet loss?

*Happens when network packets fail to reach their destination. Common Causes:*

- *Network congestion: too much traffic overloads routers/switches*
- *Weak Wi-Fi signal*
- *Hardware problems*
- *Overloaded devices*
- *Software/firewall issue*
- *Packet Corruption: fail integrity checks*
- *ISP/network outages*
- *Cyberattacks: DOS, DDoS*

Right click a packet: **Follow -> TCP Stream**

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## **Cross-System Enumeration**

Get Windows IP and Kali IP.

From Kali: ping WINDOWS\_IP

From Windows: ping KALI\_IP

Why do some systems block ICMP?

*To reduce their exposure to attackers and limit certain types of network abuse. It makes systems less visible or harder to probe.*

From Kali: nmap WINDOWS\_IP

**Nmap -Pn**

Now try: **nmap -sV WINDOWS\_IP**

What information leakage can occur from version detection?

*It can leak detailed information about the software running on a system, which helps attackers identify: vulnerabilities, exploits, OS, applications, and patch levels.*

Now: try **nmap -A WINDOWS\_IP**

What is OS fingerprinting?

*The process of identifying a target system's operating system by analyzing how it responds to network traffic.*

How does NMAP guess operating systems?

*It sends specially crafted network packets to a target and analyzes how the target responds. (OS fingerprinting). It compares the target's responses against a large database of known OS fingerprints.*

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